

```
ubuntu@ip-172-31-5-40:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-07 06:09:32 UTC; 1h 35min ago
     Docs: man:nginx(8)
  Main PID: 2223 (nginx)
    Tasks: 3 (limit: 1008)
  Memory: 3.0M (peak: 5.3M)
     CPU: 28ms
  CGroup: /system.slice/nginx.service
          └─2223 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
            └─2225 "nginx: worker process"
              └─2226 "nginx: worker process"
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.



Jan 07 06:09:52 ip-172-31-5-40 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-5-40:~\$ sudo systemctl status mysql

```
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-07 06:10:06 UTC; 1h 36min ago
 Main PID: 2995 (mysqld)
    Status: "Server is operational"
     Tasks: 39 (limit: 1008)
  Memory: 367.7M (peak: 387.0M)
     CPU: 36.596s
    CGroup: /system.slice/mysql.service
            └─2995 /usr/sbin/mysqld
```

Jan 07 06:10:04 ip-172-31-5-40 systemd[1]: Starting mysql.service - MySQL Community Server...

Jan 07 06:10:06 ip-172-31-5-40 systemd[1]: Started mysql.service - MySQL Community Server.

ubuntu@ip-172-31-5-40:~\$ mysql -u appuser -p appdb

Enter password:

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 31

Server version: 8.0.44-0ubuntu0.24.04.2 (Ubuntu)

owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> SHOW TABLES;
```

Tables_in_appdb
users

1 row in set (0.00 sec)

```
mysql> SELECT * FROM users;
```

id	name	email
1	Alice	alice@example.com
2	Bob	bob@example.com

2 rows in set (0.00 sec)

```
mysql> exit
```

```
ubuntu@ip-172-31-5-40:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-07 06:10:06 UTC; 1h 36min ago
     Main PID: 2995 (mysqld)
    Status: "Server is operational"
      Tasks: 39 (limit: 1008)
     Memory: 367.7M (peak: 387.0M)
        CPU: 36.596s
    CGroup: /system.slice/mysql.service
            └─2995 /usr/sbin/mysqld

Jan 07 06:10:04 ip-172-31-5-40 systemd[1]: Starting mysql.service - MySQL Community Server...
Jan 07 06:10:06 ip-172-31-5-40 systemd[1]: Started mysql.service - MySQL Community Server.
ubuntu@ip-172-31-5-40:~$ mysql -u appuser -p appdb
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 31
Server version: 8.0.44-0ubuntu0.24.04.2 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW TABLES;
+-----+
| Tables_in_appdb |
+-----+
| users           |
+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM users;
+-----+-----+-----+
| id | name | email |
+-----+-----+-----+
|  1 | Alice | alice@example.com |
|  2 | Bob   | bob@example.com   |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```

```
ubuntu@ip-172-31-5-40:~$ sudo -u postgres psql pgdb
psql (16.11 (Ubuntu 16.11-0ubuntu0.24.04.1))
Type "help" for help.
```

```
pgdb=# \dt
SELECT * FROM customers;
Did not find any relation named "SELECT".
\dt: extra argument "*" ignored
\dt: extra argument "FROM" ignored
\dt: extra argument "customers;" ignored
pgdb=# SELECT * FROM customers;
 id | name  | email
----+-----+-----
  1 | Charlie | charlie@example.com
  2 | Diana  | diana@example.com
(2 rows)
```

```
pgdb=#
pgdb=#
pgdb=# exit
ubuntu@ip-172-31-5-40:~$
```

```
ubuntu@ip-172-31-5-40:~$
ubuntu@ip-172-31-5-40:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-07 07:43:37 UTC; 5min ago
 TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
    Main PID: 22696 (dockerd)
         Tasks: 9
        Memory: 78.3M (peak: 88.5M)
          CPU: 1.707s
       CGroup: /system.slice/docker.service
               └─22696 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.403237353Z" level=info msg="worker nrr2hc3go9uv37g4rvgt0b14z was successfully registered" method=(*Dispatcher).register"
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.406862890Z" level=info msg="Initializing Libnetwork Agent" advertise-addr=172.31.5.40 data-path-addr= listen-addr=0.0.0.0 local-ad
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.408714536Z" level=info msg="New memberlist node - Node:ip-172-31-5-40 will use memberlist nodeID:617ccea0b646 with config:&{NodeID
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.419939321Z" level=info msg="initialized VXLAN UDP port to 4789 " module=node node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.421506860Z" level=info msg="Node 617ccea0b646/172.31.5.40, joined gossip cluster"
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.422567638Z" level=info msg="Node 617ccea0b646/172.31.5.40, added to nodes list"
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.430450297Z" level=info msg="cluster update event" module=dispatcher node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.680781937Z" level=info msg="sbJoin: gwep4 ''->'408cfefd46bb', gwep6 ''->'' eid=408cfefd46bb ep=gateway_ingress-sbox net=docker_gw
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.697173307Z" level=info msg="sbJoin: gwep4 ''->'408cfefd46bb', gwep6 ''->'' eid=fb5b366f4928 ep=ingress-endpoint net=ingress nid=k
Jan 07 07:48:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:48:52.423544325Z" level=info msg="NetworkDB stats ip-172-31-5-40(617ccea0b646) - netID:ker925am1lkiizy0mb0k91jen leaving:false netPeers:
ubuntu@ip-172-31-5-40:~$
```

```
ubuntu@ip-172-31-5-40:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:d4aaab6242e0cace87e2ec17a2ed3d779d18fbfd03042ea58f2995626396a274
Status: Downloaded newer image for hello-world:latest
```

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>


```
ubuntu@ip-172-31-5-40:~$ docker info | grep Swarm
```

```
Swarm: active
```

```
ubuntu@ip-172-31-5-40:~$ docker node ls
```

ID	HOSTNAME	STATUS	AVAILABILITY	MANAGER	STATUS	ENGINE	VERSION
nrr2hc3go9uv37g4rvgt0b14z *	ip-172-31-5-40	Ready	Active	Leader		29.1.3	

```
ubuntu@ip-172-31-5-40:~$ docker node inspect --pretty nrr2hc3go9uv37g4rvgt0b14z
```

```
ubuntu@ip-172-31-5-40:~$ sudo vim /etc/logrotate.d/mysql-server
ubuntu@ip-172-31-5-40:~$
ubuntu@ip-172-31-5-40:~$
ubuntu@ip-172-31-5-40:~$
ubuntu@ip-172-31-5-40:~$ sudo logrotate -f /etc/logrotate.conf
ubuntu@ip-172-31-5-40:~$ ls -lh /var/log/mysql/
total 4.0K
-rw-r----- 1 mysql adm    0 Jan  7 08:16 error.log
-rw-r----- 1 mysql adm 1.1K Jan  7 06:15 error.log.1.gz
ubuntu@ip-172-31-5-40:~$ ls -lh /var/log/postgresql/
total 4.0K
-rw-r----- 1 postgres adm    0 Jan  7 08:16 postgresql-16-main.log
-rw-r----- 1 postgres adm 2.2K Jan  7 08:16 postgresql-16-main.log.1
ubuntu@ip-172-31-5-40:~$ sudo systemctl status mysql
```

```
ubuntu@ip-172-31-5-40:~$ sudo ls -lh /var/log/postgresql/
total 4.0K
-rw-r----- 1 postgres adm 2.2K Jan  7 07:36 postgresql-16-main.log
ubuntu@ip-172-31-5-40:~$
```

```
Last login: Wed Jan  7 07:44:44 2026 from 152.57.152.109
ubuntu@ip-172-31-5-40:~$ sudo ls -lh /var/log/nginx/
total 8.0K
-rw-r----- 1 www-data adm 3.9K Jan  7 07:43 access.log
-rw-r----- 1 www-data adm  76 Jan  7 06:09 error.log
```

```
Jan 07 07:43:36 ip-172-31-5-40 systemd[1]: Starting docker.service - Docker Application Container Engine...
Jan 07 07:43:36 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:36.853380144Z" level=info msg="Starting up"
Jan 07 07:43:36 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:36.858252714Z" level=info msg="OTEL tracing is not configured, using no-op tracer provider"
Jan 07 07:43:36 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:36.859369581Z" level=info msg="CDI directory does not exist, skipping: failed to monitor for changes: no such file or directory" dir=/var/p
Jan 07 07:43:36 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:36.859391545Z" level=info msg="CDI directory does not exist, skipping: failed to monitor for changes: no such file or directory" dir=/etc/c
Jan 07 07:43:36 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:36.861366672Z" level=info msg="detected 127.0.0.53 nameserver, assuming system-resolved, so using resolv.conf: /run/systemd/resolve/resolv
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.013959289Z" level=info msg="Creating a containerd client" address=/run/containerd/containerd.sock timeout=1m0s
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.035640385Z" level=info msg="Loading containers: start."
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.035700783Z" level=info msg="Starting daemon with containerd snapshotter integration enabled"
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.065607770Z" level=info msg="Restoring containers: start."
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.145688844Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.151723125Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.466685709Z" level=info msg="Loading containers: done."
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.502604451Z" level=info msg="Docker daemon" commit=fbf3ed2 containerd-snapshotter=true storage-driver=overlays version=29.1.3
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.502762533Z" level=info msg="Initializing buildkit"
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.599183062Z" level=info msg="Completed buildkit initialization"
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.621921732Z" level=info msg="Daemon has completed initialization"
Jan 07 07:43:37 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:37.621984195Z" level=info msg="API listen on /run/docker.sock"
Jan 07 07:43:37 ip-172-31-5-40 systemd[1]: Started docker.service - Docker Application Container Engine.
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.155213761Z" level=info msg="Listening for connections" addr="[:]:2377" module=node node.id=nrr2hc3go9uv37g4rvgt0b14z proto=tcp
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.171669703Z" level=info msg="Listening for local connections" addr=/var/run/docker/swarm/control.sock module=node node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.160558063Z" level=info msg="33014f0ac7f83868 switched to configuration voters=()" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.165819724Z" level=info msg="33014f0ac7f83868 became follower at term 0" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.166596678Z" level=info msg="newRaft 33014f0ac7f83868 [peers: [], term: 0, commit: 0, applied: 0, lastindex: 0, lastterm: 0]" module=raft
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.166625011Z" level=info msg="33014f0ac7f83868 became follower at term 1" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.171669703Z" level=info msg="33014f0ac7f83868 switched to configuration voters=(3675305678634236008)" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.179327995Z" level=info msg="33014f0ac7f83868 switched to configuration voters=(3675305678634236008)" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.179381064Z" level=info msg="33014f0ac7f83868 is starting a new election at term 1" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.179403212Z" level=info msg="33014f0ac7f83868 became candidate at term 2" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.180550979Z" level=info msg="33014f0ac7f83868 received MsgVoteResp from 33014f0ac7f83868 at term 2" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.181793488Z" level=info msg="33014f0ac7f83868 has received 1 MsgVoteResp votes and 0 vote rejections" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.181826823Z" level=info msg="33014f0ac7f83868 became leader at term 2" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.181836682Z" level=info msg="raft.node: 33014f0ac7f83868 elected leader 33014f0ac7f83868 at term 2" module=raft node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.186402672Z" level=info msg="Creating default ingress network" module=node node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.191311633Z" level=info msg="leadership changed from not yet part of a raft cluster to nrr2hc3go9uv37g4rvgt0b14z" module=node node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.191563135Z" level=info msg="dispatcher starting" module=dispatcher node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.362731725Z" level=info msg="manager selected by agent for new session: { }" module=node/agent node.id=nrr2hc3go9uv37g4rvgt0b14z
Jan 07 07:43:52 ip-172-31-5-40 dockerd[22696]: time="2026-01-07T07:43:52.362871727Z" level=info msg="waiting 0s before registering session" module=node/agent node.id=nrr2hc3go9uv37g4rvgt0b14z
```